Chapter 3. Probabilistic Regression, Regularization and Kernel Ridge

Linear regression from a probabilistic view, regularization, ridge regression, the kernel trick

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1 Gaussian Likelihood Explains Squared Error

Motivation. Why squared error? If the noise is Gaussian, maximizing the likelihood is exactly minimizing the sum of squares. Each loss in the course corresponds in the same way to a noise model.

Theorem 3.1: MLE under Gaussian noise is least squares

If $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, then

$$\log p(\mathbf{y} \mid X, \boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2.$$

For fixed σ^2, maximizing the likelihood is minimizing the SSE, so $\hat{\boldsymbol{\beta}}_{\text{MLE}}$ solves the normal equations. Maximizing over σ^2 then gives $\hat{\sigma}^2 = \text{SSE}/n$.

Proof. The density of y_i is $(2\pi\sigma^2)^{-1/2}\exp\{-(y_i - \mathbf{x}_i^\top\boldsymbol{\beta})^2/(2\sigma^2)\}$; independence multiplies densities and the log turns the product into the displayed sum. Differentiating in σ^2: $-\frac{n}{2\sigma^2} + \frac{\text{SSE}}{2\sigma^4} = 0$. ■

2 Ridge Regression

Motivation. Least squares is unstable when $X^\top X$ is nearly singular or there are more features than samples. Adding $\lambda\|\mathbf{w}\|^2$ makes the problem well posed, shrinks the poorly determined directions most, and trades a little bias for a large reduction in variance.

2.1 Definition and closed form

Definition 3.2: ridge

For $\lambda > 0$, $J(\boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{\lambda}{2}\|\boldsymbol{\beta}\|^2$.

Theorem 3.3: ridge solution

$\nabla J = X^\top X\boldsymbol{\beta} - X^\top\mathbf{y} + \lambda\boldsymbol{\beta}$ and the unique minimizer is $\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda\mathbf{I})^{-1}X^\top\mathbf{y}$, even when X is rank deficient.

Proof. $\mathbf{v}^\top(X^\top X + \lambda\mathbf{I})\mathbf{v} = \|X\mathbf{v}\|^2 + \lambda\|\mathbf{v}\|^2 > 0$ for $\mathbf{v} \neq \mathbf{0}$, so the matrix is positive definite, invertible, and J is strictly convex. ■

2.2 MAP interpretation

With prior $\boldsymbol{\beta} \sim N(\mathbf{0}, \tau^2\mathbf{I})$, $-\log p(\boldsymbol{\beta} \mid \mathbf{y}) = \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{1}{2\tau^2}\|\boldsymbol{\beta}\|^2 + \text{const}$; multiplying by σ^2 gives ridge with $\lambda = \sigma^2/\tau^2$. A tight prior (small τ) means strong shrinkage.

2.3 Shrinkage in singular directions

Let $X = U\Sigma V^\top$ (thin SVD, singular values s_i). Since $X^\top X + \lambda\mathbf{I} = V(\Sigma^2 + \lambda\mathbf{I})V^\top$,

$$\hat{\boldsymbol{\beta}}_\lambda = \sum_i \frac{s_i}{s_i^2 + \lambda}(\mathbf{u}_i^\top\mathbf{y})\mathbf{v}_i, \qquad X\hat{\boldsymbol{\beta}}_\lambda = \sum_i \frac{s_i^2}{s_i^2 + \lambda}(\mathbf{u}_i^\top\mathbf{y})\mathbf{u}_i.$$

Least squares fits $\sum_i (\mathbf{u}_i^\top \mathbf{y})\mathbf{u}_i$; ridge multiplies the ith component by $s_i^2/(s_i^2+\lambda) \in (0,1)$. Small singular values (directions along which the data barely vary, where least squares amplifies noise by $1/s_i$) are shrunk most. $\lambda \to 0$ recovers least squares; $\lambda \to \infty$ sends coefficients to zero.

2.4 Bias and variance of ridge

Assume $\mathbf{y} = X\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$ with $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$, and X of full column rank with $X^\top X = VDV^\top$, $D = \mathrm{diag}(d_1, \dots, d_p)$, $d_i > 0$.

Theorem 3.4: ridge trades bias for variance

(i) $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}_0 = -\lambda(X^\top X + \lambda\mathbf{I})^{-1}\boldsymbol{\beta}_0$; for $\lambda > 0$ the estimator is unbiased iff $\boldsymbol{\beta}_0 = \mathbf{0}$.

(ii) $\mathrm{Cov}(\hat{\boldsymbol{\beta}}_\lambda) = \sigma^2(X^\top X + \lambda\mathbf{I})^{-1}X^\top X(X^\top X + \lambda\mathbf{I})^{-1}$ and

$$\operatorname{tr}\mathrm{Cov}(\hat{\boldsymbol{\beta}}_\lambda) = \sigma^2 \sum_i \frac{d_i}{(d_i+\lambda)^2} < \sigma^2 \sum_i \frac{1}{d_i} = \operatorname{tr}\mathrm{Cov}(\hat{\boldsymbol{\beta}}_0) \qquad (\lambda > 0).$$

(iii) $\lambda \mapsto \|\hat{\boldsymbol{\beta}}_\lambda\|$ is decreasing.

Proof. (i) $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda\mathbf{I})^{-1}X^\top X\boldsymbol{\beta}_0$. From $(X^\top X + \lambda\mathbf{I})^{-1}(X^\top X + \lambda\mathbf{I}) = \mathbf{I}$ we get $(X^\top X + \lambda\mathbf{I})^{-1}X^\top X = \mathbf{I} - \lambda(X^\top X + \lambda\mathbf{I})^{-1}$, which gives the bias. The matrix $(X^\top X + \lambda\mathbf{I})^{-1}$ is invertible, so the bias vanishes iff $\boldsymbol{\beta}_0 = \mathbf{0}$.

(ii) $\hat{\boldsymbol{\beta}}_\lambda = M\mathbf{y}$ with $M = (X^\top X + \lambda\mathbf{I})^{-1}X^\top$, so $\mathrm{Cov} = M(\sigma^2\mathbf{I})M^\top$. All three matrices are diagonalized by V, giving eigenvalues $\sigma^2 d_i/(d_i+\lambda)^2$; the trace is their sum, and $(d_i+\lambda)^2 > d_i^2$.

(iii) $\|\hat{\boldsymbol{\beta}}_\lambda\|^2 = \sum_i s_i^2(\mathbf{u}_i^\top\mathbf{y})^2/(s_i^2+\lambda)^2$ by orthonormality of $\mathbf{v}_i$; each term decreases in λ. ∎

3 Other Penalties and the Lasso

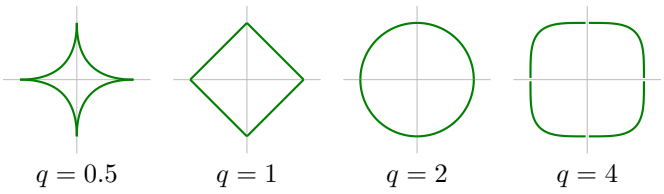
More generally one penalizes $\sum_j |\beta_j|^q$. $q = 2$ is ridge (round level sets, pure shrinkage); $q = 1$ is the lasso, whose diamond-shaped level sets have corners on the axes and produce exact zeros; $0 < q < 1$ is nonconvex.

Theorem 3.5: soft thresholding

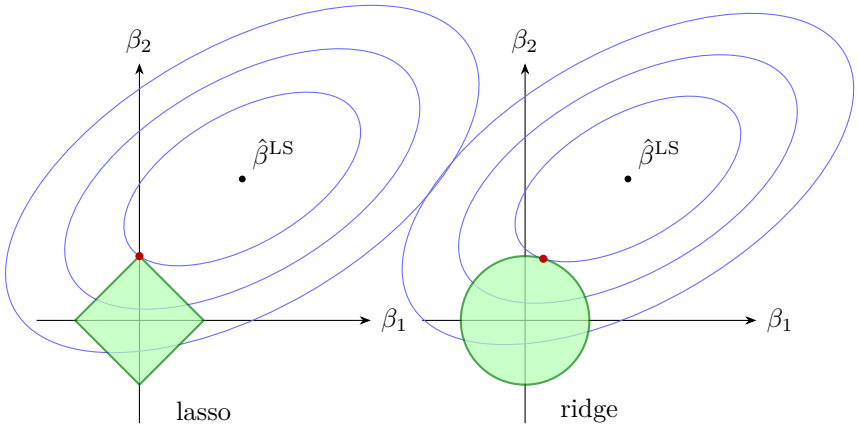
$\arg\min_b \ \frac{1}{2}(b-z)^2 + \lambda|b| = \mathrm{sign}(z)\max(|z|-\lambda, 0)$, whereas $\arg\min_b \frac{1}{2}(b-z)^2 + \frac{\lambda}{2}b^2 = z/(1+\lambda)$.

Proof. The objective is strictly convex; the optimality condition is $0 \in b - z + \lambda\,\partial|b|$, where $\partial|b| = \{\mathrm{sign}\, b\}$ for $b \neq 0$ and $[-1,1]$ at 0. If $b > 0$: $b = z - \lambda$, valid iff $z > \lambda$. If $b < 0$: $b = z + \lambda$, valid iff $z < -\lambda$. If $b = 0$: need $z \in \lambda[-1,1]$, i.e. $|z| \le \lambda$. The ridge case is the stationarity condition $b - z + \lambda b = 0$. ∎

Lasso moves every coefficient by a fixed amount and cuts small ones to zero (sparsity); ridge rescales proportionally and never produces exact zeros.



Unit balls $\sum_j |\beta_j|^q \le 1$ of the penalties: $q = 1$ is the lasso, $q = 2$ ridge.



Constrained least squares: the smallest RSS contour meets the lasso diamond at a corner (a zero coefficient) and the ridge disk at a generic point.

4 Kernel Ridge Regression

Motivation. Nonlinear regression can be done by mapping inputs to a feature space and fitting ridge there. The solution depends on the data only through inner products, so a kernel lets us work with infinitely many features at the cost of an $n \times n$ linear system.

4.1 Feature-space and sample-space forms

Let $\boldsymbol{\phi}(\mathbf{x}) \in \mathbb{R}^m$ be a feature map and, $\Phi = [\boldsymbol{\phi}(\mathbf{x}_1) \cdots \boldsymbol{\phi}(\mathbf{x}_n)] \in \mathbb{R}^{m \times n}$ (columns are samples). Ridge in feature space minimizes $J = \frac{1}{2}\|\mathbf{y} - \Phi^\top \mathbf{w}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$, so

$$\nabla J = -\Phi\mathbf{y} + (\lambda \mathbf{I}_m + \Phi\Phi^\top)\mathbf{w}, \qquad \hat{\mathbf{w}} = (\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi\mathbf{y}.$$

Theorem 3.6: push-through identity

$(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi = \Phi(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$. Hence with the Gram matrix $K = \Phi^\top\Phi$, $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j) = \boldsymbol{\phi}(\mathbf{x}_i)^\top \boldsymbol{\phi}(\mathbf{x}_j)$,

$$\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha}, \quad \boldsymbol{\alpha} = (K + \lambda \mathbf{I}_n)^{-1}\mathbf{y}, \qquad f(\mathbf{x}) = \boldsymbol{\phi}(\mathbf{x})^\top \hat{\mathbf{w}} = \sum_{i=1}^{n} \alpha_i k(\mathbf{x}_i, \mathbf{x}) = \mathbf{k}_{\mathbf{x}}^\top (K + \lambda \mathbf{I}_n)^{-1}\mathbf{y}.$$

Proof. $\Phi(\lambda \mathbf{I}_n + \Phi^\top\Phi) = \lambda\Phi + \Phi\Phi^\top\Phi = (\lambda \mathbf{I}_m + \Phi\Phi^\top)\Phi$. Multiply on the left by $(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}$ and on the right by $(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$ (both positive definite). The two identity matrices have different sizes. ∎

Remark 3.7: the kernel trick and cost

$\boldsymbol{\phi}$ appears only through inner products, so only the kernel k is needed. The primal form solves an $m \times m$ system and the dual an $n \times n$ one: the kernel form wins when $m \gg n$ (e.g. the Gaussian kernel, whose feature space is infinite-dimensional). Merely mapping to higher dimensions is feature engineering; computing through k without forming $\boldsymbol{\phi}$ is the kernel trick. An alternative derivation substitutes $\mathbf{w} = \Phi\boldsymbol{\alpha}$ and obtains $K[(K + \lambda\mathbf{I})\boldsymbol{\alpha} - \mathbf{y}] = \mathbf{0}$; when K is singular one may not cancel K, but the displayed $\boldsymbol{\alpha}$ is a solution and all solutions give the same $\hat{\mathbf{w}}$.

4.2 Kernels

- Linear: $k(\mathbf{x}, \mathbf{z}) = \mathbf{x}^\top \mathbf{z}$.
- Quadratic in 2D: $(\mathbf{x}^\top \mathbf{z})^2 = x_1^2 z_1^2 + x_2^2 z_2^2 + 2x_1 x_2 z_1 z_2$, so $\boldsymbol{\phi}(\mathbf{x}) = (x_1^2, x_2^2, \sqrt{2}x_1 x_2)$. In 1D, $(1 + xz)^2$ has $\boldsymbol{\phi}(x) = (1, \sqrt{2}x, x^2)$.
- Gaussian: $k(\mathbf{x}, \mathbf{z}) = \exp(-\|\mathbf{x} - \mathbf{z}\|^2/(2\sigma^2))$; σ is the bandwidth.
- Normalized: $\tilde{k}(\mathbf{x}, \mathbf{z}) = k(\mathbf{x}, \mathbf{z})/\sqrt{k(\mathbf{x}, \mathbf{x})k(\mathbf{z}, \mathbf{z})}$.

Theorem 3.8: Gram matrices are PSD

For any feature-map kernel and any $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a}^\top K\mathbf{a} = \|\sum_i a_i \boldsymbol{\phi}(\mathbf{x}_i)\|^2 \geq 0$.

Not every similarity function is a valid kernel; positive semidefiniteness of all Gram matrices is the requirement.

4.3 Worked examples

Example 3.9: linear kernel

$\mathbf{x}_1 = (1, 2)$, $\mathbf{x}_2 = (2, 0)$, $\mathbf{y} = (3, 1)$, $\lambda = 1$: $K = \begin{pmatrix} 5 & 2 \\ 2 & 4 \end{pmatrix}$, $(K + \mathbf{I})^{-1} = \frac{1}{26}\begin{pmatrix} 5 & -2 \\ -2 & 6 \end{pmatrix}$, $\boldsymbol{\alpha} = (1/2, 0)$, $\hat{\mathbf{w}} = \frac{1}{2}\mathbf{x}_1 = (1/2, 1)$. At $\mathbf{x} = (1, 0)$: $\mathbf{k}_\mathbf{x} = (1, 2)$ and $f(\mathbf{x}) = 1/2 = \hat{\mathbf{w}}^\top \mathbf{x}$.

Example 3.10: quadratic kernel

$\mathbf{x}_1 = (1, 1)$, $\mathbf{x}_2 = (2, 0)$, $\mathbf{y} = (3, 2)$, $k = (\mathbf{x}^\top \mathbf{z})^2$, $\lambda = 1$. Then $K_{11} = 4$, $K_{12} = 4$, $K_{22} = 16$,

$$(K + \mathbf{I})^{-1} = \frac{1}{69}\begin{pmatrix} 17 & -4 \\ -4 & 5 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \frac{1}{69}\begin{pmatrix} 43 \\ -2 \end{pmatrix}.$$

At $\mathbf{x}_t = (1, 0)$, $\mathbf{k}_{\mathbf{x}_t} = (1, 4)$ and $f(\mathbf{x}_t) = (43 - 8)/69 = 35/69 \approx 0.507$. Check in feature space: $\boldsymbol{\phi}(\mathbf{x}_1) = (1, 1, \sqrt{2})$, $\boldsymbol{\phi}(\mathbf{x}_2) = (4, 0, 0)$, $\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha} = (35, 43, 43\sqrt{2})/69$ and $\boldsymbol{\phi}(\mathbf{x}_t) = (1, 0, 0)$ give the same $35/69$.